
A High-Frequency Nowcasting Framework Using Google Trends, Mixed-Frequency Models, and Machine Learning

Ahmad Ilu

`aii2400012.pec@buk.edu.ng`

May 15, 2026

Outline

1. Executive Snapshot
2. GDP Growth & Nowcasting
3. Economic Sentiment Index (ESI)
4. Inflation Diagnostics
5. Monetary Policy Stance
6. Exchange Rate & External Sector
7. Policy Outlook & Conclusions

Disclaimer

The views, interpretations, and conclusions expressed in this presentation are solely those of the author and do not necessarily reflect the views of any affiliated institutions, organizations, or individuals.

All data points, numerical values, and scenarios described herein are purely hypothetical and intended exclusively for educational and illustrative purposes.

This presentation does not constitute professional advice, official policy, or factual representation of real-world events.

Executive Snapshot: Key Macro Indicators at a Glance

[GDP] Real Output — Q4 2025 (NBS)

YoY Growth (Q4-25)	4.07% ▲
Full-Year 2025	3.87%
Oil Sector	6.79%
Non-Oil Sector	3.99%
Economy Size (Nom.)	₦ 441.5 Trillion

[MPR] Monetary Policy — 304th MPC

MPR	26.5% (↓50 bps)
CRR (Commercial)	45.0%
CRR (Merchant)	16.0%
Liquidity Ratio	30.0%
SDF Corridor	+50 / -450 bps

[CPI] Inflation — April 2026 (NBS)

Headline YoY	15.69% ▼ (vs 26.82% Apr-25)
Core Inflation	15.86%
Food Inflation	16.06%
MoM Headline	2.13%

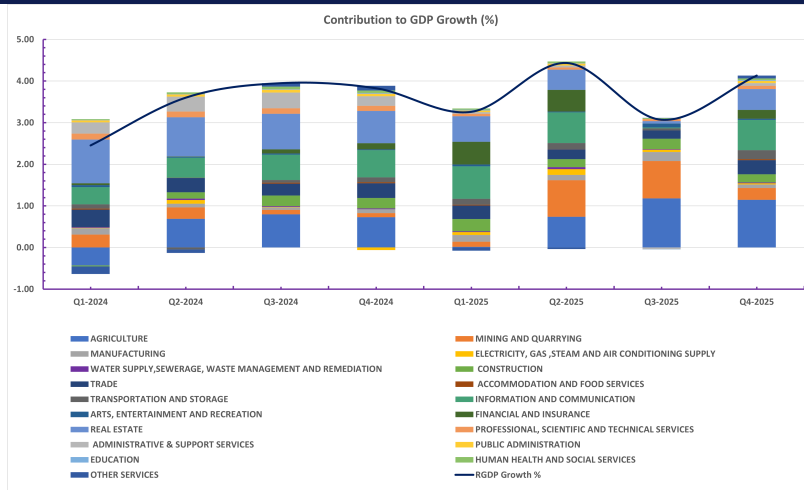
[FX] Exchange Rate — May 15, 2026

USD/NGN Official	₦ 1,343.77
USD/NGN Market	₦ 1,400
EUR/NGN	₦ 1,585
GBP/NGN	₦ 1,870

Key Signal for the 305th MPC (May 20, 2026)

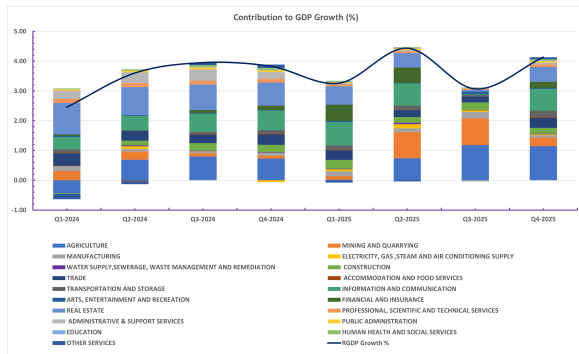
Inflation re-accelerating for 2 consecutive months (Mar–Apr 2026) while real GDP momentum holds above 4%.

Real GDP Growth: Historical Trajectory (2022–2025)



Note: GDP rebased (Base year: 2019 — NBS). Full-year 2025: **3.87%** vs **3.38%** in 2024 (+49bps YoY improvement). Q4-25 is *only the second time in a decade* (ex. post-COVID) that quarterly growth exceeds 4.0%. *Source: NBS Quarterly GDP Reports.*

GDP Sectoral Decomposition: Q4 2025



Key Growth Drivers

- **Finance & Insurance: 8.30%** — banking reform dividend; capital market deepening
- **ICT: 7.55%** — digital economy shift; telecoms and fintech
- **Oil Sector: 6.79%** — crude at **1.58m bpd**, up from 1.54m bpd (Q4-24)
- **Construction: 5.08%** — infrastructure pipeline
- **Agriculture: 4.00%** — improved security in food belts

Nominal Economy

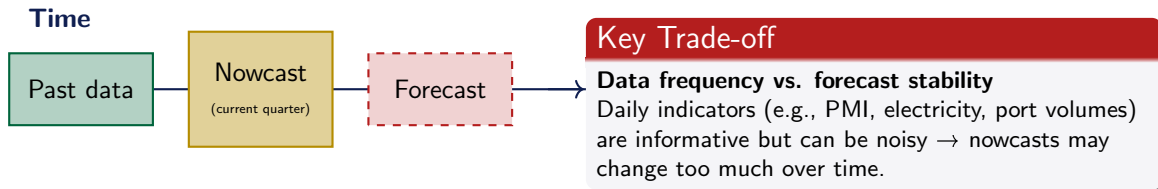
Economy: ~~₦441.5tn~~ (2025) vs ₦372.8tn (2024).
Agriculture: **25.67%** of nominal GDP.

Structural Weakness

Manufacturing: only **1.13% YoY** — below Q4-24 and Q3-25 levels. High energy costs, tight credit, and FX input costs continue to constrain industrial base.

What is Nowcasting?

Real-time assessment of the current economy



Definition

“Nowcast” = now + forecast. Real-time prediction of a low-frequency variable (e.g., quarterly GDP) using higher-frequency data (daily, weekly, monthly).

Why policy-makers need nowcasts

Assess the **current state** of the economy in real-time, when only incomplete information is available (GDP is released with a long lag).

References:

Lewis, D. J., Mertens, K., & Stock, J. H. (2020). *Nowcasting GDP*.
Woloszko, N. (2020). *Tracking activity in real time with Google Trends*. OECD Economics Dept.

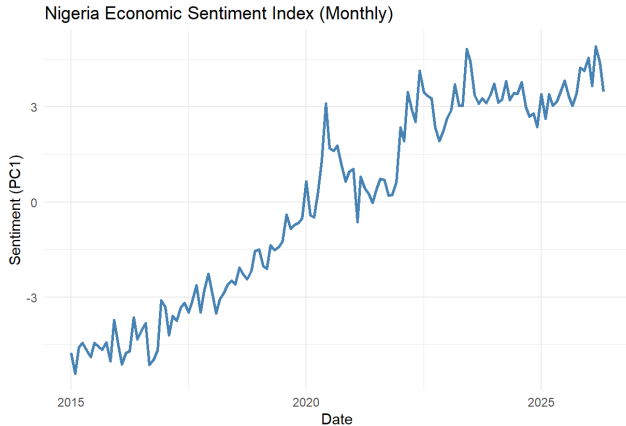
Introducing the Economic Sentiment Index (ESI)

A real-time, high-frequency measure of Nigeria's economic mood using **Google Trends search data**.

- ▶ **Data:** 50+ search terms (inflation, dollar to naira, flight to Nigeria, flight tickets, Abuja, Lagos, Kano, rice, PMS, LPG, Geopolitics, fuel subsidy, CBN, Dangote, Tinubu, minimum wage, etc.) monthly, 2015–present.
- ▶ **Method:** Principal Component Analysis (PCA) extracts the first principal component = ESI.
- ▶ **Scale:** Mean = 0, SD = 1. Positive values = above-average sentiment (relative to 2015–2026 baseline).

ESI ↑ → **stronger economic optimism** → **leading indicator for GDP**

ESI Trajectory (2015–2026)



Source: Author's Computation

Period	ESI signal	Key context
2015–2019	−4 to −1	Oil crash, 2016 recession
2020–2022	0 to +3	COVID rebound, FX/inflation
2023–2025	> +3	Reforms, Dangote refinery
2026 Q1	+4.37	Peak confidence

Google Trends Search Intensity in Nigeria

Word size & color = average search interest (2015–present)



Using the ESI Alongside Traditional Indicators

Complement, not substitute – what the ESI adds

- ▶ **Real-time pulse:** Available monthly, while GDP is quarterly with a lag.
- ▶ **Leads sentiment:** Captures household/business attention before official data.
- ▶ **Validates other nowcasts:** ESI nowcast aligns with the range from our main macro model.

Important Caveats

- ▶ ESI measures **search behaviour**, not transactions – a leading but noisy indicator.
- ▶ “Positive” means better than the **2015–2026 average**, not absolute good.
- ▶ Political/seasonal events (elections, holidays) can create temporary spikes.

Integrate ESI with PMI, credit, and FX data for a richer nowcasting toolkit.

GDP Nowcasting: Q1 2026 Projection

Nowcasting Framework (Mixed-Frequency)

Applying an **Bridge Equation** / **MIDAS** and **Machine Learning** approach using:

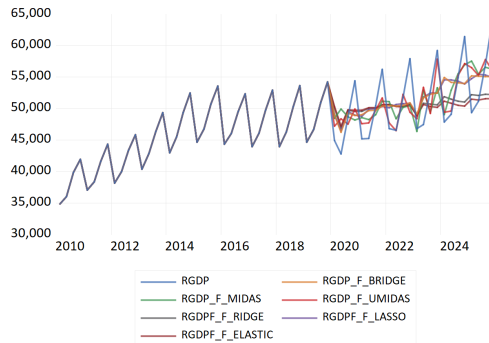
- ▶ **Activity**: PMI, NESG Business Confidence Monitor (BCM)
- ▶ **Financial**: Credit to Private Sector (CPS)
- ▶ **Google Trends**: Economic Sentiment Index (ESI)

Q1 2026 Nowcast Range

3.80% — 4.20%

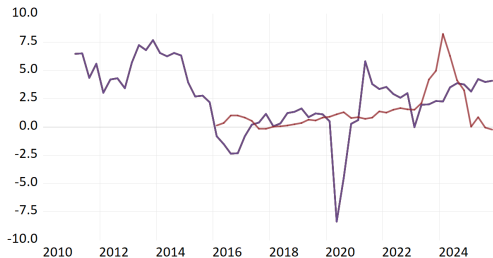
Point Estimate: 4.0% YoY

Central Scenario | Uncertainty: $\pm 0.3\text{pp}$



Nowcast Result: Q1 2026

Combining ESI, regularised models, and traditional indicators



Model	Q1-26 nowcast (YoY %)	Plausibility
RIDGE	3.62%	✓ Conservative
ELASTIC NET	4.64%	✓ Optimistic
Simple mean	4.64%	✓
Selected nowcast	4.0%	Central

Why not MIDAS (6.22%) or BRIDGE (14.2%)?

Linear models assume the same relationship holds when the Economic Sentiment Index reaches an all-time high (+4.37). In reality, the relationship flattens – extreme optimism does not translate into extreme growth.

Implication for MPC

Growth above potential (≈ 3.8 – 4.2%) → supports **HOLD**. No overheating – sentiment-driven growth is not inflationary.

Nowcast Model Evaluation

In-sample performance (2010Q1 – 2025Q4)

Metric	BRIDGE	MIDAS	RIDGE	LASSO	ELASTIC	Simple mean
RMSE (₦ bn)	2,909	2,766	3,151	2,907	3,259	2,897
MAPE (%)	2.99	2.76	3.07	2.96	3.10	2.96
Encompassing p-value	0.14	0.90	0.003	0.14	0.0006	–

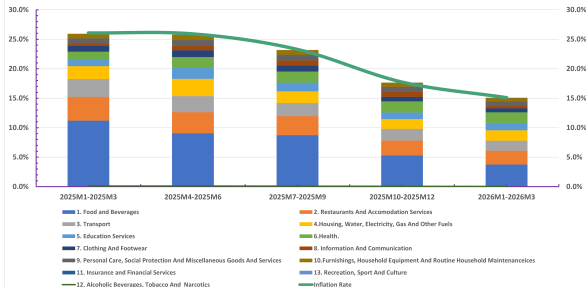
Key findings

- ▶ **MIDAS** has the lowest error and best encompassing test → best in-sample performer.
- ▶ **But** MIDAS extrapolates poorly when ESI is outside historical range (Q1-26 ESI = +4.37).
- ▶ **RIDGE / ELASTIC** produce plausible Q1-26 nowcasts because they shrink extreme coefficients.

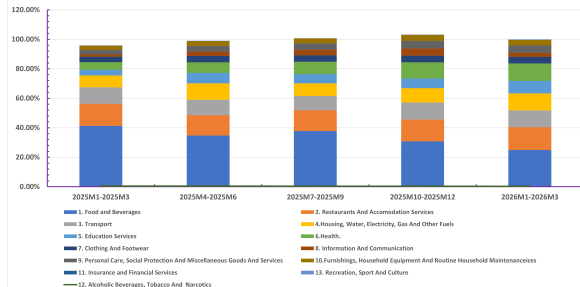
Based on 68 quarterly observations, 6 forecasting models.

Absolute and Relative Contribution to Inflation (Apr 2025 – Apr 2026)

Absolute Contributions to Inflation



Relative Contributions to Inflation



Inflation Drivers & the CBN Dual-Mandate Tension

April 2026: CPI Contribution by Division

Division	Contribution (pp)
Food & Non-Alcoholic Beverages	6.40
Restaurants & Accommodation	3.56
Transport	1.70
Miscellaneous Goods & Services	≈1.20
All other divisions	≈2.83
Headline (YoY)	15.69%

Source:

NBS CPI Report, May 15, 2026.

Disinflation Interrupted

- ▶ An **11-month disinflation streak** ended in March 2026
- ▶ **Two consecutive upticks:** 15.06% (Feb) → 15.38% (Mar) → **15.69% (Apr)**
- ▶ **Food inflation resurging:** 8.89% (Jan) → 12.12% (Feb) → 14.31% (Mar) → **16.06% (Apr)**
- ▶ **Core MoM eased sharply:** 1.03% — underlying demand softening
- ▶ Rural (16.36%) vs Urban (15.40%) gap signals a **supply-side food shock**, not demand-pull

New Sub-Indices (YoY, Apr 2026)

Farm Produce	19.8%
Services	16.7%
Goods	15.7%

Regional Price Dispersion

Hotspots: Sokoto **25.74%**, Bauchi **22.52%**, Zamfara **22.03%**

Anchors: Edo **5.91%**, Borno **6.72%**, Jigawa **7.04%**

Northern states under persistent cost-push pressure; food

Hybrid Inflation Expectations

Model Specification

$$\pi_t^e = \alpha\pi_t + (1 - \alpha)\pi_t^*$$

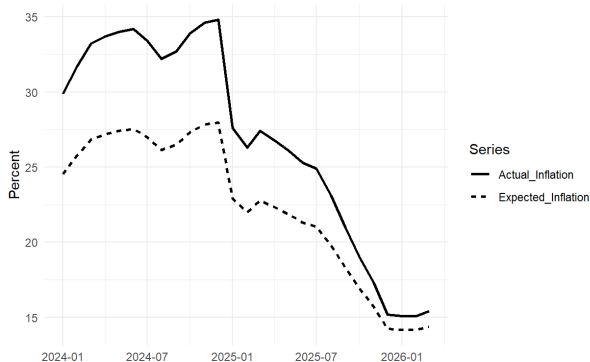
Where:

- ▶ π_t^e : Expected inflation
- ▶ π_t : Actual inflation
- ▶ π_t^* : Policy anchor (12%)
- ▶ α : Weight on actual inflation (0.7)

Interpretation

- ▶ Hybrid expectations: adaptive + policy credibility
- ▶ 70% driven by current inflation
- ▶ 30% anchored to policy target
- ▶ Captures persistence and gradual anchoring

Nigeria: Actual vs Hybrid Inflation Expectations



Source: Author's Computation

Used in Phillips Curve, monetary policy rule, and forecasting.

Monetary Policy Transmission & 305th MPC Decision Matrix

Current Policy Parameters (304th MPC, Feb 2026)

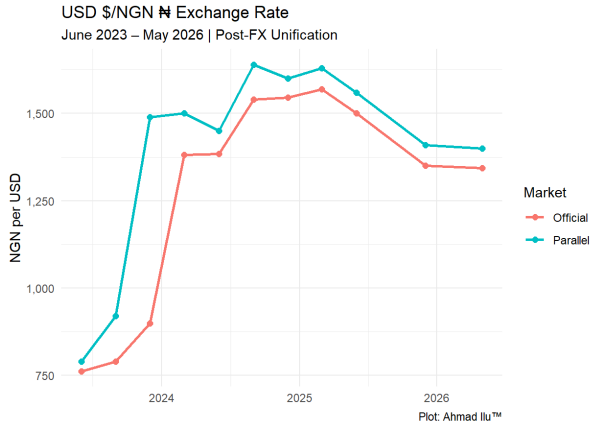
Instrument	Level / Setting
MPR (Benchmark Rate)	26.50%
Standing Lending Facility	27.00% (+50bps)
Standing Deposit Facility	22.00% (−450bps)
CRR — Commercial Banks	45.00%
CRR — Merchant Banks	16.00%
CRR — Non-TSA Public Dep.	75.00%
Liquidity Ratio	30.00%

Corridor design: The asymmetric +50/−450bps corridor *penalises idle SDF deposits* while capping SLF access. CBN intent: channel liquidity toward private sector credit.

305th MPC Scenario Analysis

Decision	Key Rationale
HOLD (Base, 65%)	Back-to-back inflation uptick; credibility cost of premature easing; real rates positive
−25 bps (Dovish, 30%)	Core MoM softening; FX stable; above-potential growth; measured easing manageable
+25 bps (Hawkish, 5%)	Pre-emptive against food shock; global oil spike; Naira vulnerability

Naira Exchange Rate: Trajectory & Post-Unification Stabilisation



FX Chronology & Key Metrics

- ▶ **Jun 2023:** FX windows unified; Naira floated from ~~≈₦460~~ to ~~≈₦800~~ overnight
- ▶ **Peak weakness:** ~~≈₦1,640~~ (Sep 2024)
- ▶ **Stabilisation:** CBN interventions, remittances, improved oil receipts
- ▶ **May 2026:** Official ~~₦1,343.77~~ — Market ~~₦1,400~~
- ▶ **Spread:** ~~≈₦56 (4.2%)~~ — narrow by post-float standards

External Tailwinds

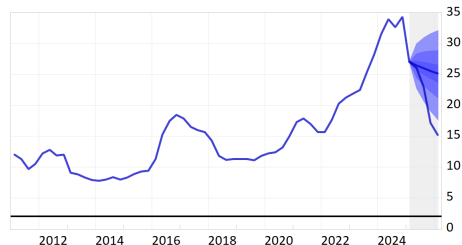
- ▶ Crude output: **1.58m bpd** (Q4-25), recovering from 2024 lows
- ▶ FTSE Russell: Nigeria upgraded to **Frontier Market** (Apr 2026); FPI inflows improving
- ▶ High MPR sustaining carry-trade; Naira demand

External Sector: Inflation Linkage & Policy Implications

FX Stability as Inflation Anchor

- ▶ Naira stabilisation has been the **primary disinflation channel** since Q4 2024
- ▶ **Imported Food inflation: 10.5%** (Apr-26) $\Rightarrow$ well below headline; FX pass-through contained
- ▶ **Energy inflation: only 4.6%** YoY $\Rightarrow$ domestic fuel price stability a positive outlier
- ▶ High MPR (26.5%) attracts **FPI carry** $\Rightarrow$ FX supply/demand balance sustained
- ▶ Premature rate cut risks **carry unwind** $\Rightarrow$ Naira pressure $\Rightarrow$ import inflation re-acceleration

Fan chart forecast of Nigeria's CPI



External Headwinds

- ▶ **US-Iran / Strait of Hormuz tensions** $\Rightarrow$ global oil supply shock risk
- ▶ **US Fed uncertainty** $\Rightarrow$ delayed pivot could trigger EM capital outflows

Interpretation: Low energy and imported food inflation vs. elevated farm produce and services confirms the April uptick is **domestically driven** (food supply shock) — not an FX pass-through event. This supports **Hold** over a hike.

Integrated Assessment: Pre-305th MPC Scorecard

[GDP] Growth Outlook

Latest GDP (Q4-25)	4.07% YoY ▲
Full-Year 2025	3.87%
Q1-26 Nowcast	≈4.0%
IMF 2026 Forecast	3.8%–4.1%
Assessment	Resilient

[FX] External Sector

USD/NGN Official	₦1,343.77
Parallel Spread	≈4.2%
Crude Output (Q4-25)	1.58m bpd
FTSE Upgrade	Frontier Market
Assessment	Stable

[CPI] Inflation Outlook

Headline (Apr-26)	15.69% YoY ▼
Core (Apr-26)	15.86%
Food Trend	Re-accelerating ▼
MoM Core	1.03% (easing)
Assessment	Caution

Monetary Policy

MPR (current)	26.5%
Real MPR (approx.)	≈+10.8pp
Easing Phase Started	Sep 2025
Cumulative Cuts	100 bps
305th Call	HOLD

Key Risks, Upside Catalysts & Analytical Framework

Downside Risks

- ❶ **Food inflation resurgence** — seasonal drought, food belt insecurity
- ❷ **Oil production disruption** — NNPC risks, Niger Delta unrest
- ❸ **Global energy shock** — US-Iran / Hormuz; fuel import spiral
- ❹ **Fiscal dominance risk** — election-cycle spending pressures
- ❺ **EM capital flow reversal** — if US Fed delays pivot
- ❻ **Credit squeeze** — CRR at 45% limits private investment credit

Upside Catalysts

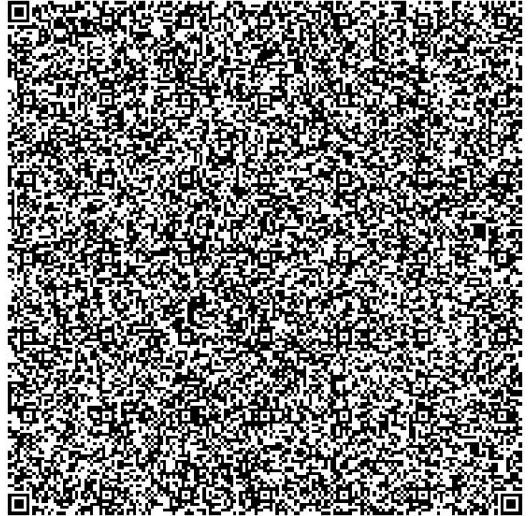
- ❶ **FTSE Frontier upgrade** — sustained FPI inflows; Naira support
- ❷ **Dangote Refinery ramp-up** — FX demand reduction; domestic fuel
- ❸ **Agricultural productivity** — improved security + dry-season farming
- ❹ **ICT & fintech growth** — structural GDP diversification
- ❺ **Diaspora remittances** — strong NGN incentive for formal inflows
- ❻ **Oil price rally** — fiscal windfall; FX reserve accumulation

Nigeria Macroeconomic Diagnostics

"Applying rigorous macroeconomic diagnostics to inform evidence-based monetary policy discourse in Nigeria."

Bottom Line: Nigeria's macro fundamentals are improving; GDP above 4%, FX stable, disinflation structurally underway. However, back-to-back inflation upticks in March and April 2026 counsel against rushing the easing cycle. **The 305th MPC should Hold at 26.5%**, signal data-dependency, and retain the option for a further –25bps cut in Q3-26 if food supply pressures ease and inflation resumes its downward trend.

Scan to save my Contact



Thank you for Reading!